AI-Based Electrical Load Forecasting System

Capstone Project Report | Second-Year EEE

Abstract

This project demonstrates a machine-learning pipeline for one-hour-ahead electrical load forecasting. Historical readings are cleaned, resampled to hourly averages, transformed into time and lag features, and used to train a Random Forest regression model. Chronological splitting holds out later observations for testing.

1. Problem Statement

Estimate the next hour's average active power (kW) from historical electricity consumption. Forecasts can support planning and energy monitoring.

2. Dataset

Recommended final dataset: UCI Individual Household Electric Power Consumption. It contains 2,075,259 one-minute measurements from one household in Sceaux, France (December 2006–November 2010). The target field, Global_active_power, is measured in kW. Missing values are represented by '?'. Dataset: https://archive.ics.uci.edu/dataset/235/individual+household+electric+power+consumption. DOI: 10.24432/C58K54.

Important artefact note

The model, charts, and metrics bundled with this ZIP are an offline synthetic demonstration, not actual meter data. Run `python train_model.py --uci` to download and train on the public dataset before submitting final results.

3. Tools

Python, Pandas, NumPy, Matplotlib, Scikit-learn, Joblib, and Streamlit.

4. Methodology

(1) Load and sort timestamped readings. (2) Convert '?' to missing values and clean invalid readings. (3) Resample one-minute readings to hourly mean kW. (4) Engineer hour, weekday, month, weekend, lag and rolling-mean features. (5) Shift the target one hour ahead. (6) Use the first 80% for training and the last 20% for testing, without shuffling. (7) Train a 150-tree Random Forest. (8) Report MAE, RMSE, R^2 and persistence-baseline MAE.

5. Initial Demo Results

These results are calculated from the included synthetic dataset only. MAE = 0.0800 kW; RMSE = 0.1011 kW; R^2 = 0.8842; persistence baseline MAE = 0.1171 kW. Do not claim these are UCI dataset results.

6. Evaluation Metrics

MAE is the average absolute error in kW. RMSE is the square root of mean squared error and penalizes larger errors. R^2 compares model predictions to a mean-prediction baseline; it can be negative. The persistence baseline predicts that the next hour equals the latest observed load.

7. Outputs

The pipeline creates a daily load profile, actual-versus-predicted chart, feature-importance chart, test predictions CSV, JSON metrics, and a serialized model.

8. Limitations

The UCI data represents one French household and may not generalize to an Indian building. Weather, occupancy, holidays and equipment status are not included. This is a one-hour-ahead forecasting prototype, not a safety-critical controller.

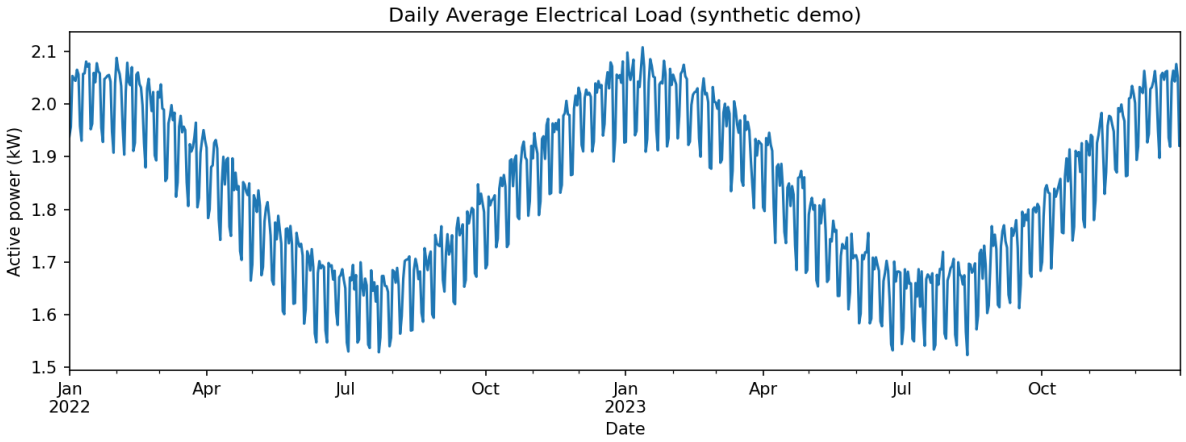
9. Conclusion

The project demonstrates an end-to-end workflow from data preparation to machine-learning evaluation and dashboarding. Before submission, run the model on the UCI dataset and update the final results using outputs/metrics.json.

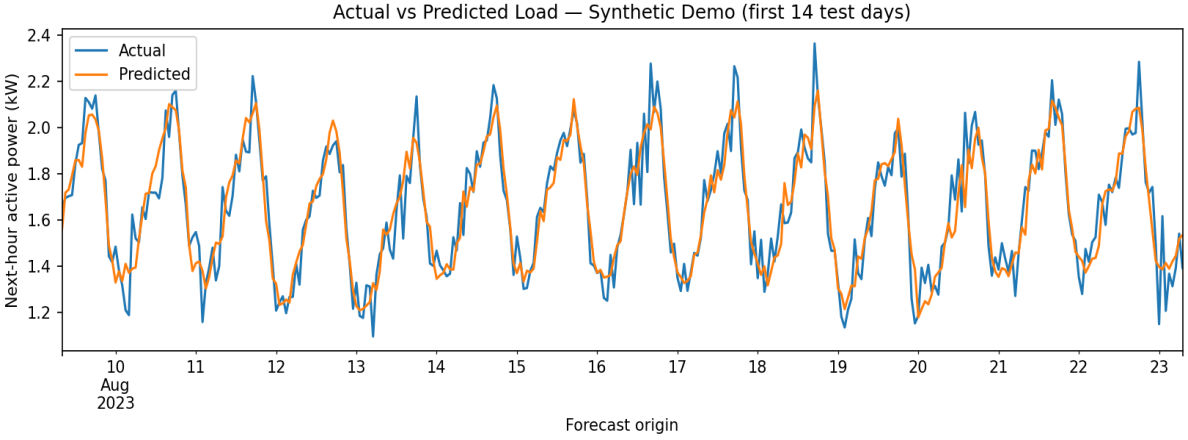
References

Hebrail, G. & Berard, A. (2006). Individual Household Electric Power Consumption. UCI Machine Learning Repository. DOI: 10.24432/C58K54. Streamlit documentation: https://docs.streamlit.io/get-started/fundamentals/main-concepts.

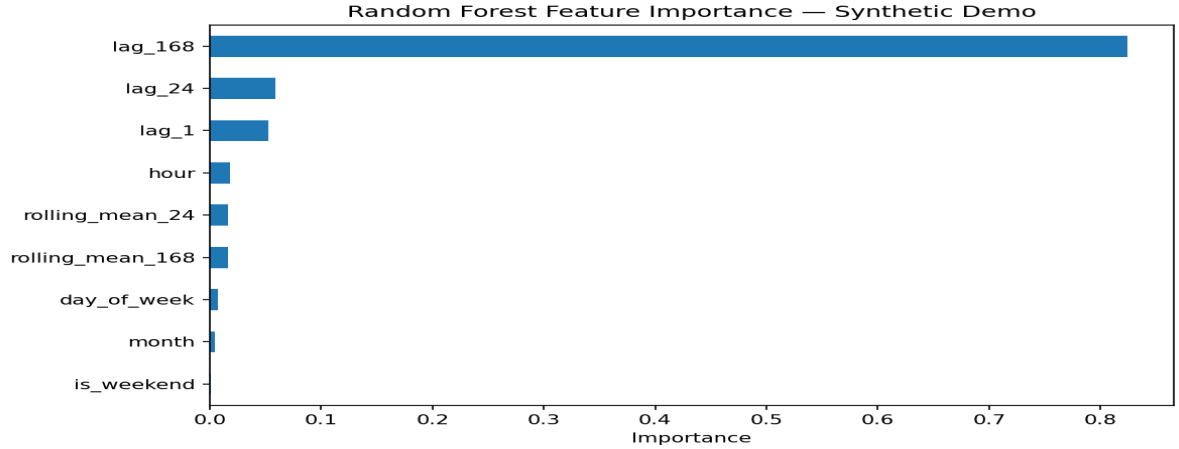
Daily average load profile (synthetic demo)



Actual vs predicted load (synthetic demo)



Feature importance (synthetic demo)


Random Forest Feature Importance — Synthetic Demo
lag_168
lag_24
lag_1
hour
rolling_mean_24
rolling_mean_168
day_of_week
month
is_weekend
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Importance